1. Explain the purpose of the Forget Gate in LSTM. Explain the formula:

   $f_t = \sigma(W_f [h_{t-1}, x_t] + b_f)$

2. Explain the role of the Input Gate in LSTM. Explain the formulas:

   $i_t = \sigma(W_i [h_{t-1}, x_t] + b_i)$

   $\tilde{C}_t = \tanh(W_c [h_{t-1}, x_t] + b_c)$

3. Explain how the cell state is updated in LSTM using the formula:

   $C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$

4. Explain the purpose of the Output Gate in LSTM. Explain the formulas:

   $o_t = \sigma(W_o [h_{t-1}, x_t] + b_o)$

   $h_t = o_t * \tanh(C_t)$

5. Why are three gates used in LSTM? Explain the responsibilities of Forget Gate, Input Gate, and Output Gate with formulas.
6. Differentiate between Forget Gate, Input Gate, and Output Gate in terms of functionality and formula.
7. Explain how Forget Gate controls old memory and Input Gate controls new memory in LSTM.
8. Explain how the Output Gate determines the hidden state in LSTM.
9. “Forget Gate removes unnecessary information, Input Gate adds important information, and Output Gate controls the final output.”
   Explain this statement with formulas.
10. Explain the complete workflow of an LSTM cell using the formulas of Forget Gate, Input Gate, cell state update, and Output Gate.